



Megapack 2 XL Specification

Revision 1.8

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PRODUCT SPECIFICATIONS

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1 System Description.....	2
1.1 Applications.....	2
1.2 Capacity Maintenance.....	2
1.3 Constituent Parts.....	2
1.4 Definitions.....	3
1.5 Standards and Regulations.....	4
2 Electrical Specifications.....	5
2.1 Power and Energy.....	5
2.1.1 Scaling Power and Energy.....	5
2.2 Round-Trip Efficiency and Energy Consumption	5
2.3 Charge and Discharge Limitations.....	9
2.4 Inverter Specifications.....	10
2.4.1 AC Interface.....	10
2.4.1.1 Medium Voltage Transformer Connection.....	10
2.4.1.2 AC Breaker.....	11
2.4.1.3 Voltage Ride Through.....	11
2.4.1.4 Frequency Ride Through.....	12
2.4.1.5 Anti-Islanding Features.....	12
2.4.1.6 Inverter Configurations.....	13
2.4.1.7 Battery Module Configurations.....	15
3 Environmental Specifications.....	17
3.1 Ambient Temperature.....	17
3.2 Elevation.....	17
3.3 Seismic, Shock, and Vibration.....	17
3.4 Wind.....	17
3.5 Roof Loading.....	18
3.6 Solar Loading.....	18
4 Mechanical Specifications.....	19
4.1 Dimensions and Mass.....	19
4.2 Transportation.....	19
4.3 Installation.....	19
4.4 Enclosure.....	20
4.5 Audible Noise.....	20
5 Communication and Control.....	21
Revision History.....	22



1 System Description

Tesla Megapack 2 XL (Megapack) is an all-in-one utility-scale energy storage system optimized for cost and performance. It scales to the space, power, and energy requirements of any site from 1.3 MWh to 1 GWh+. The product includes a custom enclosure, batteries, bi-directional inverter, and thermal system. At the site level, a Tesla Site Controller with intelligent software allows the coordination and operation of the Megapack.

This turnkey system is designed to maximize savings and prolong battery life. Megapacks have the most advanced battery technology and dispatch optimization software to quickly learn and predict a facility's energy patterns. Tesla's proprietary storage dispatch software can charge and discharge autonomously to maximize customer value.

Megapack's architecture consists of battery cells aggregated into modules, and power electronics modules. Both are thermally managed by an integrated liquid cooling and heating system for thermal safety, enhanced performance, and reliability. The architecture also includes sensors supporting embedded monitoring and controls as well as electrical interface equipment including an industry standard breaker for AC protection.

1.1 Applications

Megapack is designed for utilities and large-scale commercial projects. Our team of experts will help you design a system that meets your project goals and maximizes your site potential.

Megapack applications include:

- **Renewable Smoothing** – Balance the flow of renewables to the grid by storing and discharging energy
- **Demand Support** – Discharge energy during peak demand to support distribution infrastructure
- **Infrastructure Investment** – Postpone costly grid infrastructure upgrades by storing power at a single location
- **Voltage & Frequency Regulation** – Stabilize voltage levels by absorbing reactive power and adjusting output
- **Market Participation** – Provide energy support to the grid in response to system operator alerts
- **Microgrid** – Build a localized grid that can disconnect from the main power grid

1.2 Capacity Maintenance

Megapacks can be augmented with a Capacity Maintenance Agreement (CMA), which guarantees power and energy capacity over the life of the product. A capacity-maintained system requires the addition of battery modules over time to maintain power and energy for up to 20 years. Technical details and characteristics differ between a product with Capacity Maintenance Agreement and a product without. These differences are highlighted in the relevant sections within this specification. Capacity maintenance is only offered for projects with a minimum size of 10 MWh.

1.3 Constituent Parts

Each Megapack system includes the following components provided by Tesla:

- Smart inverter
- AC main breaker
- Battery modules
- Thermal system



- Low-voltage customer interface panel
- Tesla Site Controller

1.4 Definitions

AC-rated real power (kW) – Nominal nameplate AC real power rating of the system. Throughout Megapack documentation, the term *real power* may be used synonymously with the term *active power*.

Apparent power (kVA) – Nominal nameplate AC apparent power rating of the system.

Battery cell – The smallest non-divisible energy component of the Megapack, assembled into a battery module in series and parallel arrays.

Battery module – A field-replaceable unit that integrates battery cells, fusing, and battery management system functions. The battery module interfaces are output electrical connections, thermal interface, and communication connections.

CMA – Capacity Maintenance Agreement, used to define a custom power and energy guarantee over the life of the system.

Efficiency degradation (%) – Represented with a stated efficiency value at the beginning of life and the minimum seen over 10 years and 365 cycles/year.

Fully operational – The operating condition where the system is capable of discharging or charging at full power.

Heat mode – A software control feature of the Megapack that is used to regulate battery cell temperature in order to maintain AC-rated real power and/or energy at low ambient temperatures.

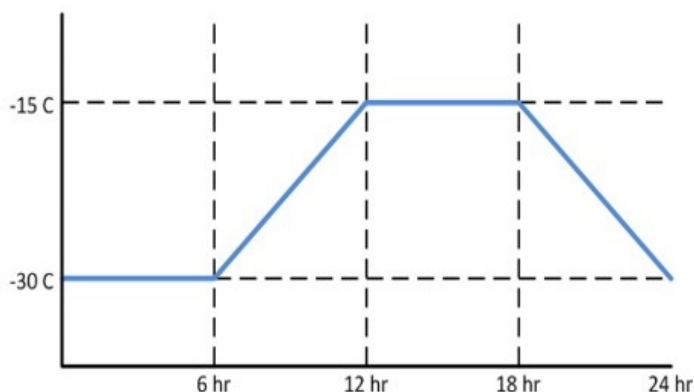
Inverter or PCS – Bi-directional power conversion system that couples each Megapack with the power grid (AC power).

Megapack – The complete AC-coupled system, inclusive of enclosure, battery modules, inverter, and thermal system.

Megapack energy consumption – Self-discharge and energy consumption of the Megapack to power its thermal system and internal auxiliary loads.

Profile_{COLD} – The ambient temperature daily profile for cold weather standby use, indicative of winter in Grand Forks, ND, is defined below:

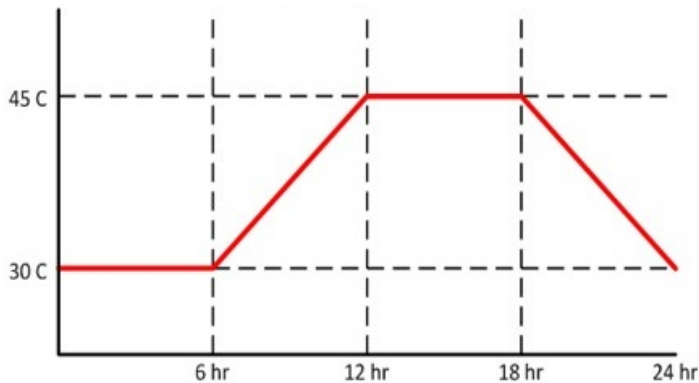
Figure 1. Cold Daily Temperature Profile





Profile_{HOT} – The ambient temperature daily profile for hot weather standby use, indicative of summer in Palm Springs, CA, is defined below:

Figure 2. Hot Daily Temperature Profile



Round-trip efficiency (RTE, %) – Defined as charging of the system from 0% SOE to 100% SOE at AC-rated real power, immediately followed by discharging of the system from 100% SOE to 0% SOE at AC-rated real power. The round-trip AC-AC energy efficiency shall be measured at the AC terminals of the inverter and shall include parasitic loads.

SOE with CMA – Battery State of Energy: Available energy / Rated energy (0-100%).

SOE without CMA – Battery State of Energy: Available energy / Full Megapack energy (0-100%).

STC – Standard Test Conditions, defined as the system soaked at 25°C and 1 atmosphere (101.3 kPa) of pressure.

System duration – The amount of time the Megapack can continuously charge or discharge power at AC-rated real power when new.

T_{AMBCOLD} – The ambient temperature for cold weather performance is defined as the system soaked at -20°C. This temperature definition is used to define the system performance in a nominally cold temperature climate. This temperature does not define the operating limitation or temperature rating of the system.

T_{AMBHOT} – The ambient temperature for hot weather performance is defined as the system soaked at 45°C. This temperature definition is used to define the system performance in a nominally hot temperature climate. This temperature does not define the operating limitation or temperature rating of the system.

Tesla Site Controller – Site-level control interface that is the single point of feedback and control for a physical installation. It is also responsible for collecting feedback data from each individual Megapack, running algorithms to optimize the system operation, and providing commands to inverters.

1.5 Standards and Regulations

The system and components are or will be compliant with the standards and directives listed in the *Megapack 2 XL Compliance Packet*, to be found on the Tesla Partner Portal at <https://partners.tesla.com/>.



2 Electrical Specifications

Megapack is rated in terms of net-delivered power and minimum energy at its AC output terminals. Loads and losses, including power conversion, thermal system losses, auxiliary loads, and chemical/ionic losses are considered internal to the system and ratings are net of these loads. Losses between the AC output terminals and the point of interconnection with the customer/utility are site-dependent and excluded from the rated power and energy.

2.1 Power and Energy

System ratings are defined in kW and kWh as measured at the AC terminals of the Megapack. The given energy ratings (kWh) in this document are minimum initial energy ratings.

Megapack is capable of providing AC minimum initial energy (kWh) at AC-rated real power (kW) when discharged from 100% SOE at STC. Megapack is capable of charging or discharging at AC-rated real power for the life of the system. The maximum duration of charge/discharge will decrease over the life of the system. The kW is net of thermal loads and can be achieved throughout life.

A fully populated Megapack has the following nominal ratings at 480 V AC:

Table 1. Nominal Fully Populated Ratings

System Duration	Max. Configurable AC Apparent Power (kVA)	AC-Rated Real Power (kW)	AC Minimum Initial Energy (kWh)
2-hour	2400.0	1927.2	3854.4
4-hour	1512.0*	979.2	3916.8

* The apparent power rating of the inverter is expected to increase in 2023 manufacturing from 63 kVA to 68 kVA. This will increase the total Max. Configurable AC Apparent Power to 1632.0 kVA. To understand how this changes system configuration capability, refer to [Battery Module Configurations on page 15](#).

2.1.1 Scaling Power and Energy

Megapack can be requested with lower factory-configured inverter ratings. See [Inverter Configurations on page 13](#) for more details.

Megapack can be requested with fewer battery modules. See [Battery Module Configurations on page 15](#) for more details.

2.2 Round-Trip Efficiency and Energy Consumption

2.2.1 Round-Trip Efficiency

Round-trip efficiency (RTE) is defined as the relative difference in energy charged to the system from 0% SOE to 100% SOE at AC-rated real power, immediately followed by energy discharged of the system from 100% SOE to 0% SOE at AC-rated real power. The quotient of energy discharged over energy charged is the RTE.

The round-trip efficiency, inclusive of thermal system load and all internal auxiliary loads during the cycle, is specified in the table below:

**Table 2. Round-Trip Efficiency**

Parameter	2-Hour System	4-Hour System
Minimum Round-Trip Efficiency at Year 0	STC: 92.0% T_{AMBHOT} : 90.5% $T_{AMBCOLD}$: 92.0%	STC: 93.5% T_{AMBHOT} : 92.5% $T_{AMBCOLD}$: 93.0%
Round-Trip Efficiency at Year 15*	STC: 90.0% T_{AMBHOT} : 88.5% $T_{AMBCOLD}$: 89.5%	STC: 92.0% T_{AMBHOT} : 90.5% $T_{AMBCOLD}$: 91.5%

* Indicative figures. Exact value will depend on battery utilization and climate over the 15 years.

Indicative round-trip efficiency (year 0 and year 10) under STC for a single Megapack at different power levels is shown in the graphs below.

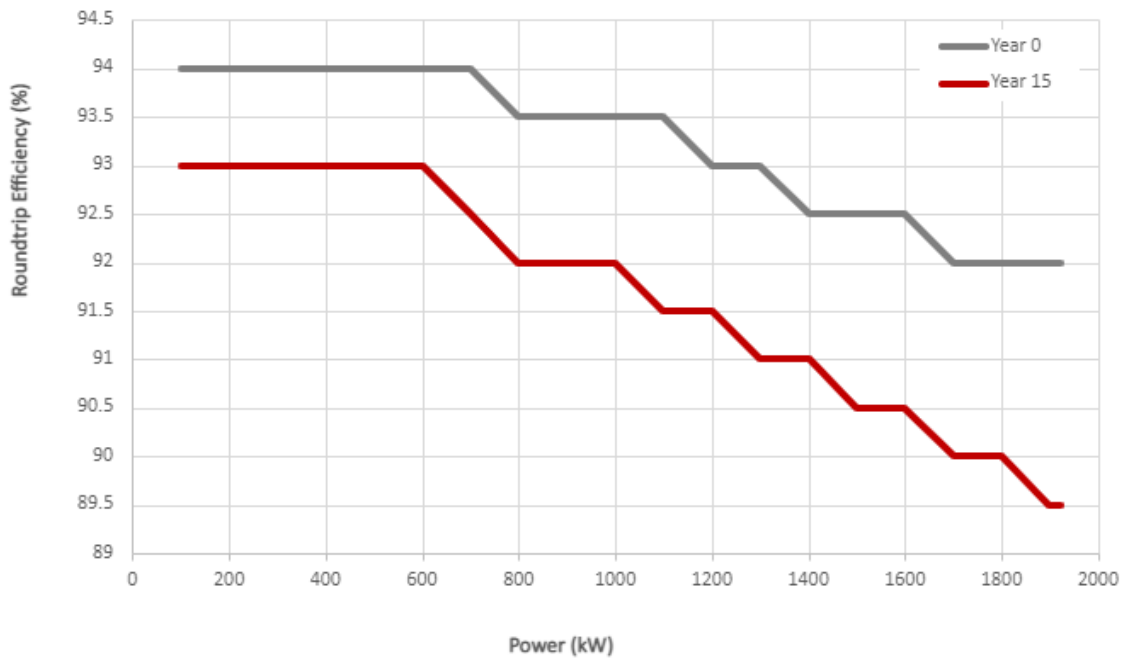
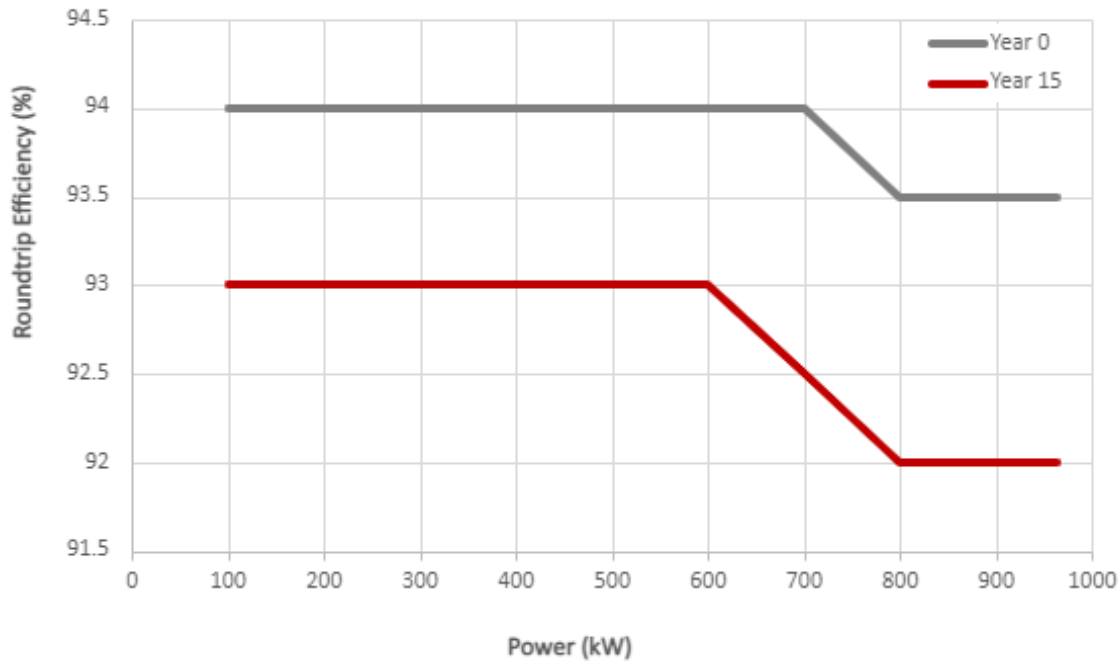
Figure 3. Round-Trip Efficiency: 2-hour Megapack


Figure 4. Round-Trip Efficiency: 4-hour Megapack


2.2.2 Energy Consumption



NOTE: All specifications below assume heat mode is not enabled by the user. Additional energy consumption due to heat mode is detailed in [Charge and Discharge Limitations on page 9](#).

When Megapack is not actively charging or discharging, the Megapack may still self-discharge and consume energy, which is referred to as Megapack energy consumption (kWh).

Megapack energy consumption (kWh) was evaluated for the following two operating states:

- **Active:** The system will remain in Active state and keep its inverters powered on. Commanding the Megapack to remain in Active mode will result in increased energy consumption due to inverter switching.
- **Standby:** When the system does not receive any command within five minutes, the system will automatically move to Standby and power down its inverters to reduce energy consumption.

Pre-condition is defined as how the system was operated immediately before the 24-hour period during which Megapack energy consumption is calculated:

- **No cycle:** The system was not charging or discharging power.
- **One full cycle:** The system was charged from 0% SOE to 100% SOE at AC-rated real power (kW) and was immediately discharged from 100% SOE to 0% at AC-rated real power (kW).

Megapack energy consumption over a 24-hour period is detailed in the tables for various scenarios below.

Table 3. Megapack Energy Consumption, Standby, No Cycle

System Duration	Ambient Conditions	Energy Consumption [kWh/day]
4-Hour	STC	23
4-Hour	T _{AMBCOLD}	17



System Duration	Ambient Conditions	Energy Consumption [kWh/day]
4-Hour	T_{AMBHOT}	77
2-Hour	STC	25
2-Hour	$T_{AMBCOLD}$	17
2-Hour	T_{AMBHOT}	81

Table 4. Megapack Energy Consumption, Standby, One Full Cycle

System Duration	Ambient Conditions	Energy Consumption [kWh/day]
4-Hour	STC	63
4-Hour	$T_{AMBCOLD}$	21
4-Hour	T_{AMBHOT}	122
2-Hour	STC	67
2-Hour	$T_{AMBCOLD}$	19
2-Hour	T_{AMBHOT}	173

Table 5. Megapack Energy Consumption, Active, No Cycle

System Duration	Ambient Conditions	Energy Consumption [kWh/day]
4-Hour	STC	154
4-Hour	$T_{AMBCOLD}$	153
4-Hour	T_{AMBHOT}	208
2-Hour	STC	287
2-Hour	$T_{AMBCOLD}$	287
2-Hour	T_{AMBHOT}	343

Table 6. Megapack Energy Consumption, Active, One Full Cycle

System Duration	Ambient Conditions	Energy Consumption [kWh/day]
4-Hour	STC	197
4-Hour	$T_{AMBCOLD}$	192
4-Hour	T_{AMBHOT}	256
2-Hour	STC	338
2-Hour	$T_{AMBCOLD}$	327
2-Hour	T_{AMBHOT}	443

Finally, in addition to the Megapack energy consumption, the Tesla Site Controller has the following energy consumption per unit over a 24-hour period:

Table 7. Auxiliary Load per Tesla Site Controller

Parameter	2-Hour System Duration	4-Hour System Duration
Auxiliary Load for Tesla Site Controller [kWh/day]	0.5	0.5



2.3 Charge and Discharge Limitations

Megapack's thermal management system may be used to maintain the battery's charge and discharge capabilities across the system's rated ambient temperature range. In cold ambient conditions, operating in this state results in lower overall efficiency due to increased parasitic loads.

By activating heat mode, the 2-hour or 4-hour Megapack can be preconditioned from the following starting temperatures to allow rated charge power:

Table 8. Charge Power

Starting Temperature	Time to Full Charge Power Capability	Preheat Energy Consumption to Full Charge Power Capability	24-Hr Heating Energy Consumption for Maintaining Full Charge Power Capability
-30°C	9 hours	328 kWh	445 kWh
-20°C	6 hours	218 kWh	334 kWh
-10°C	3 hours	115 kWh	222 kWh
0°C	0 hours	0 kWh	0 kWh
10°C	0 hours	0 kWh	0 kWh
20°C	0 hours	0 kWh	0 kWh
30°C	0 hours	0 kWh	0 kWh

Similarly, the 2-hour or 4-hour Megapack can be preconditioned to allow rated discharge power:

Table 9. Discharge Power

Starting Temperature	Time to Full Discharge Power Capability	Preheat Energy Consumption to Full Discharge Power Capability	24-Hr Heating Energy Consumption for Maintaining Full Discharge Power Capability
-30°C	4 hours	152 kWh	269 kWh
-20°C	2 hours	66 kWh	173 kWh
-10°C	0 hours	0 kWh	0 kWh
0°C	0 hours	0 kWh	0 kWh
10°C	0 hours	0 kWh	0 kWh
20°C	0 hours	0 kWh	0 kWh
30°C	0 hours	0 kWh	0 kWh

Finally, the 2-hour or 4-hour Megapack can be preconditioned from the following starting temperatures to maximize the discharge energy:

Table 10. Discharge Energy

Starting Temperature	Time to Full Discharge Energy Capability	Preheat Energy Consumption to Full Discharge Energy Capability	24-Hr Heating Energy Consumption for Maintaining Full Discharge Energy Capability
-30°C	19 hours	730 kWh	768 kWh
-20°C	15 hours	563 kWh	639 kWh



Starting Temperature	Time to Full Discharge Energy Capability	Preheat Energy Consumption to Full Discharge Energy Capability	24-Hr Heating Energy Consumption for Maintaining Full Discharge Energy Capability
-10°C	11 hours	420 kWh	541 kWh
0°C	8 hours	294 kWh	409 kWh
10°C	5 hours	180 kWh	302 kWh
20°C	2 hours	70 kWh	169 kWh
30°C	0 hours	0 kWh	0 kWh

2.4 Inverter Specifications

Table 11. Inverter Specifications

Max Continuous Output Current	Factory-configurable (See Inverter Configurations on page 13)
Overload Capability	120% of rated current (10 sec max)
Nominal AC Voltage	480 V
Output Voltage Range	422-552 V AC
Nominal Frequency (configurable)	50 or 60 Hz
Frequency Range	45-66 Hz
Phases	3
System Configuration	3-wire, Wye Note: Grounded Wye required at transformer secondary
Full Load Efficiency	98.3%
California Energy Commission Weighted Efficiency	98.5%
Power Factor at Full Load	> 99%
Adjustable Power Factor (Controller Feature)	-1 to +1
Total Current Harmonic Distortion (THD)	< 5%
Power Regulation Accuracy	< 2%
Overvoltage Category	Category III up to 3000 m
Maximum Short Circuit Current Withstand	85 kAIC

2.4.1 AC Interface

2.4.1.1 Medium Voltage Transformer Connection

If Megapack is connected to a medium-voltage transformer, the transformer must be a grounded wye transformer and the connection must be made via a 3-wire circuit (3 phases, ground) connection.



It is possible to parallel multiple Megapacks on the low-voltage side of a transformer without providing additional galvanic isolation.

2.4.1.2 AC Breaker

Each Megapack includes an AC breaker with the following features:

- 85 kAIC interrupting capacity (2-hour and 4-hour Megapacks)
- GND fault detection
- Shunt trip
- Pad lockable

2.4.1.3 Voltage Ride Through

Megapack High-Voltage Ride Through (HVRT) and Low-Voltage Ride Through (LVRT) settings are adjustable and programmed during the commissioning process to meet IEEE 1547-2018 requirements and/or comply with applicable grid code. All per-unit (pu) voltages are defined with respect to 480 V AC L-L.*

Table 12. Default HVRT Settings

Voltage Range (% of Base Value)	Response Time
Category II	
$117.5 < V \leq 120$	0.2s
$115 < V \leq 117.5$	0.5s
$110 < V \leq 115$	1s
Category III	
$110 < V \leq 120$	0.083s-12s

Table 13. Default LVRT Settings

Voltage Range (% of Base Value)	Response Time
Category II	
$65 \leq V < 88$	Linear slope of 8.7s/1 p.u voltage starting at 3s @0.65 p.u
$45 \leq V < 65$	0.32s
$30 \leq V < 45$	0.16s
Category III	
$70 \leq V < 88$	20s
$50 \leq V < 70$	10s
$V < 50$	0.083s - 1s

Table 14. Setting Ranges

Parameter	Setting Range	Resolution
Voltages	0.00-Maximum HVRT Allowed	0.01 per unit
Times	0.00-60.00 sec*	0.01 sec



* Time allowed a specific per-unit voltage depends on the nominal voltage rating. Typically, default settings and acceptable ranges are tested as part of a specific certification. For deviations from certified settings in a region, contact your Tesla sales engineer for more information.

Table 15. Trip Accuracy

Trip Setting	Accuracy
Voltage	+/- 1% of nominal voltage
Frequency	+/- 0.01 Hz
Reconnection Time	1% of set point
Trip Time	.16 seconds for <5 seconds 1% of set point for >5 seconds
Active Power	+/- 5% of rated apparent power
Reactive Power	+/- 5% of rated apparent power

2.4.1.4 Frequency Ride Through

The Megapack inverter is capable of staying connected to the distribution provider's distribution or transmission system while the grid is within the frequency-time range indicated in the table below and will disconnect from the electric grid during a high or low frequency event that is outside that frequency-time range.

The Megapack inverter allows for a minimum of 3 under-frequency and 3 over-frequency trip points and times, as well as 1 under-frequency instantaneous trip and 1 over-frequency instantaneous trip.

Table 16. Frequency Ride Through

Trip Point	Frequency Range	Time (sec)	Notes
Instantaneous UF Trip	40 Hz-70 Hz	Instantaneous	0.1 Hz resolution, filtering as necessary
UF Trip Time 3	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
UF Trip Time 2	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
UF Trip Time 1	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 1	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 2	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 3	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
Instantaneous OF Trip	40 Hz-70 Hz	Instantaneous	0.1 Hz resolution, filtering as necessary

2.4.1.5 Anti-Islanding Features

The Megapack inverter includes these anti-islanding features:

- Reconnection delay timer
- Active anti-islanding: Sandia Frequency Shift implemented on all systems
- Passive anti-islanding: Configurable Rate of Change of Frequency (ROCOF) preferences

The reconnection delay timer is configurable with the following settings:

**Table 17. Reconnection Delay Timer Default Settings**

Feature Name	Effect	Setting Range	Default
Reconnect Time Delay	The amount of time Megapack waits before reconnection, after the grid returns within the Frequency and Voltage Ride Through windows defined above	0-600 sec	300 sec
Reconnect Min. Voltage	The minimum voltage at which Megapack interprets the grid is within tolerable conditions	0-150%	88.33%
Reconnect Max. Voltage	The maximum voltage at which Megapack interprets the grid is within tolerable conditions	0-150%	105.83%
Reconnect Min. Frequency	The minimum frequency at which Megapack interprets the grid is within tolerable conditions	40-70 Hz	59.3 Hz
Reconnect Max. Frequency	The maximum frequency at which Megapack interprets the grid is within tolerable conditions	40-70 Hz	60.5 Hz

Sandia Frequency Shift is enabled by default, but can be disabled if required for the application. ROCOF is configurable with the following settings:

Table 18. ROCOF Settings

Feature Name	Effect	Setting Range	Default
ROCOF Enable	Turns ROCOF on or off	n/a	Off
ROCOF Fault Limit	Sets the rate of change required for a trip	0.1-100.0 Hz/sec	1 Hz/sec
ROCOF Time Delay	Sets how long the rate of change has to be present for the inverter to trip	0-1 second	1 second

2.4.1.6 Inverter Configurations

The configurations listed below are example inverter configurations. Megapack can be requested to be factory-configured with other current limits lower than the maximum below to de-rate to meet particular project needs. Starting at the minimum factory configuration of 400 kVA, Megapack is available in 50 kVA increments up to the maximum configuration of 2400 kVA.

The Megapack inverter should be configured with a kVA rating at least 60 kVA (2-hour systems) or 25 kVA (4-hour systems) higher than the site requirement, to ensure the Megapack can supply enough AC current to simultaneously run the thermal system and discharge in accordance with commands.

Megapacks with fewer battery modules will have a lower absolute maximum, as indicated in [Battery Module Configurations on page 15](#).

Table 19. Example Inverter Configurations

Power Configuration Factory Product Code	Configured Inverter Power Rating [kVA]	Maximum Continuous Output Current [A]	Applicable Megapack System Duration
P240	2400.0	2886.8	2-Hour Maximum
P230	2300.0	2766.5	2-Hour
P220	2200.0	2646.2	2-Hour
P210	2100.0	2525.9	2-Hour
P200	2000.0	2405.6	2-Hour
P163	1632.0	1963.0	4-Hour Maximum and 2-Hour



Power Configuration Factory Product Code	Configured Inverter Power Rating [kVA]	Maximum Continuous Output Current [A]	Applicable Megapack System Duration
P140	1400.0	1683.9	4-Hour and 2-Hour
P136	1363.2	1639.4	4-Hour and 2-Hour
P130	1300.0	1563.7	4-Hour and 2-Hour
P120	1200.0	1443.4	4-Hour and 2-Hour
P110	1100.0	1323.1	4-Hour and 2-Hour
P100	1000.0	1202.8	4-Hour and 2-Hour
P040	400.0	481.1	4-Hour and 2-Hour Minimum



2.4.1.7 Battery Module Configurations

Table 20. Current Available Battery Configurations

Configuration Factory Code	Battery Module Count	2-Hour Megapack Rating			4-Hr Megapack Rating		
		AC Rated Real Power (kW)	AC Minimum Initial Energy (kWh)	Max. AC Apparent Power (kVA)	AC Rated Real Power (kW)	AC Minimum Initial Energy (kWh)	Max. AC Apparent Power (kVA)
EC08	8	642.4	1284.8	1008	326.4	1305.6	504
EC09	9	722.7	1445.4	1134	367.2	1468.8	567
EC10	10	803	1606	1260	408	1632	630
EC11	11	883.3	1766.6	1386	448.8	1795.2	693
EC12	12	963.6	1927.2	1512	489.6	1958.4	756
EC13	13	1043.9	2087.8	1638	530.4	2121.6	819
EC14	14	1124.2	2248.4	1764	571.2	2284.8	882
EC15	15	1204.5	2409	1890	612	2448	945
EC16	16	1284.8	2569.6	2016	652.8	2611.2	1008
EC17	17	1365.1	2730.2	2142	693.6	2774.4	1071
EC18	18	1445.4	2890.8	2268	734.4	2937.6	1134
EC19	19	1525.7	3051.4	2394	775.2	3100.8	1197
EC20	20	1606	3212	2400	816	3264	1260
EC21	21	1686.3	3372.6	2400	856.8	3427.2	1323
EC22	22	1766.6	3533.2	2400	897.6	3590.4	1386
EC23	23	1846.9	3693.8	2400	938.4	3753.6	1449
EC24	24	1927.2	3854.4	2400	979.2	3916.8	1512

**Table 21. Planned Available Battery Configurations: 2023**

Configuration Factory Code	Battery Module Count	2-Hour Megapack Rating			4-Hr Megapack Rating		
		AC Rated Real Power (kW)	AC Minimum Initial Energy (kWh)	Max. Configurable AC Apparent Power (kVA)	AC Rated Real Power (kW)	AC Minimum Initial Energy (kWh)	Max. Configurable AC Apparent Power (kVA)
EC08	8	642.4	1284.8	1088	326.4	1305.6	544
EC09	9	722.7	1445.4	1224	367.2	1468.8	612
EC10	10	803	1606	1360	408	1632	680
EC11	11	883.3	1766.6	1496	448.8	1795.2	748
EC12	12	963.6	1927.2	1632	489.6	1958.4	816
EC13	13	1043.9	2087.8	1768	530.4	2121.6	884
EC14	14	1124.2	2248.4	1904	571.2	2284.8	952
EC15	15	1204.5	2409	2040	612	2448	1020
EC16	16	1284.8	2569.6	2176	652.8	2611.2	1088
EC17	17	1365.1	2730.2	2312	693.6	2774.4	1156
EC18	18	1445.4	2890.8	2400	734.4	2937.6	1224
EC19	19	1525.7	3051.4	2400	775.2	3100.8	1292
EC20	20	1606	3212	2400	816	3264	1360
EC21	21	1686.3	3372.6	2400	856.8	3427.2	1428
EC22	22	1766.6	3533.2	2400	897.6	3590.4	1496
EC23	23	1846.9	3693.8	2400	938.4	3753.6	1564
EC24	24	1927.2	3854.4	2400	979.2	3916.8	1632



3 Environmental Specifications

3.1 Ambient Temperature

Megapack is capable of meeting rated power in ambient temperatures between -30°C and 50°C, and in relative humidity of up to 100% condensing.

In high altitudes, at low air pressure, the cooling capability of Megapack may be reduced, but is no worse than standard altitude-temperature de-rate as specified by standard atmosphere conditions as indicated below:

Table 22. Ambient Temperature

Altitude [m]	Maximum Ambient Temperature During Operation [°C]
0-999	50.0
1000-1999	48.5
2000-2999	42.0
3000	35.5

At no time, including during shipping, may storage conditions exceed the ranges indicated below.



NOTE: The system must be at a specified level of charge prior to 12-month storage. Contact your Tesla representative for information on long-term storage.

Table 23. Storage Conditions

Duration	Temperature Range During Storage [°C]	Maximum Humidity
Up to 12 months (365 days)	-30 to 50	Up to 100% relative humidity, condensing

3.2 Elevation

Megapack provides rated power up to 3000 m elevation above sea level in ambient temperature conditions per [Ambient Temperature on page 17](#).

3.3 Seismic, Shock, and Vibration

Megapack meets the high seismic, shock, and vibration performance levels per the following standards:

- Qualification Level - IEEE 693-2018 High PL: $ZPA=1.0$ g 5% damping
- Certification Level - ICC-ES AC 156-2018 $S_{DS}=2.50$ g $z/h=0$ $I_p=1.5$

3.4 Wind

If installed as a standalone enclosure, Megapack is able to withstand Category 5 hurricane sustained wind speeds of up to 157 mph (252 km/h).



3.5 Roof Loading

Megapack is able to withstand a snow load of up to 732.3 kilograms per square meter (150 pounds per square foot, or 7.18 kN_{Earth}/m²) on its roof or other surface.

Megapack is designed to ensure that its roof can withstand a person of up to 300 lbs (136 kg) gross weight in clearly designated areas.

3.6 Solar Loading

Megapack provides full rated power in all ambient temperature conditions ([Ambient Temperature on page 17](#)), and at maximum elevation ([Elevation on page 17](#)) under up to 1200 W/m² solar loading, in any direction.



4 Mechanical Specifications

4.1 Dimensions and Mass



NOTE: Mass as listed is maximum mass. Megapack can be configured for lighter shipping mass based on project-specific requirements.



NOTE: Enclosure dimensions are provided for nominal design guidance only. For precise dimensions, see *Megapack 2 XL Drawings* on the Partner Portal.

Table 24. Megapack Dimensions and Mass (Weight)

Width	Depth	Height	Max. Shipping Mass
8800 mm	1650 mm	2785 mm	38,100 kg
(346 ½ in)	(65 in)	(110 in)	(84,000 lb)

4.2 Transportation

Megapack is a non-divisible structure. No disassembly for transport is required.

Megapack is over-height on a normal flat rack, so for ocean shipping use of an adjustable-height flat rack (“Super Rack”) is recommended if available.

Megapack has 6 compliant ISO 1161 type lifting points for effective rigging and will be certified to AS4991-2004.

4.3 Installation



NOTE: Site design and installation guidance will be provided in the *Megapack 2 XL Design and Installation Manual*. Until available, refer to the *Megapack 2 Design and Installation Manual* for indicative guidance only.

4.3.1 Anchoring

Megapack has a total of 12 anchor brackets to be structurally affixed to a foundation. Each anchor bracket supports a total of three anchoring points, totaling 36 potential anchor points. In general, there are four types of foundations to which a Megapack may be anchored:

1. Concrete slab
2. Steel I-Beam
3. Concrete or steel pile
4. HDPE beams with ground screws

The number of anchors that will be used should be defined by a project’s Structural Engineer of Record through the site S_{DS} value and AHJ requirements.



4.3.2 Electrical Termination

Megapack is designed with two above-ground wireways through which to route AC cabling to LV transformer termination and communications cabling. The wireways do not require any site trenching construction. Each wireway is 410 mm x 101 mm (16 1/8 in x 4 in) and runs from each side of the Megapack into the conduit opening in the Customer Interface Bay. Circuits using wireways must be designed and installed in accordance with local code.

4.4 Enclosure

4.4.1 Ingress Protection

The Megapack thermal system has an ingress protection rating of IP20, while the enclosure has an ingress protection rating of IP66 and is classified to NEMA 3R certification (including rain and sprinkler test immunity).

4.4.2 Impact Protection

The Megapack enclosure has a minimum IK rating of IK09 for impact protection.

4.4.3 Salt and Fog

The Megapack enclosure is able to withstand over 1,000 hours of salt fog application per a C5M system.

4.4.4 Corrosion Resistance and Paint

The paint system of the Megapack enclosure is compliant with ISO 12944: C5I (industrial) and C5M (coastal) standards.

4.5 Audible Noise

The audible noise of Megapack, measured 10 meters from any side surface of the enclosure, is less than 75 dBA (SPL sound pressure) at full thermal system performance.



5 Communication and Control

5.1 Tesla Site Controller

The Tesla Site Controller is the single point of interface for the utility, network operator, or customer SCADA systems to control and monitor the entire energy storage site. It hosts the control algorithm that dictates the charge and discharge functions of the battery system units, aggregating real-time information and using the information to optimize the commands sent to each individual battery unit.

The battery system communicates with the Tesla Site Controller over a private TCP network via Modbus, DNP3, or REST API interfaces.

Refer to the *Tesla Energy Controls and Communications Manual* for details on the following topics:

- Available battery operational data points and definitions
- Available control modes and associated setpoints
- Protocols supported for monitoring and controls

5.2 Powerhub

Powerhub is a cloud-based monitoring platform that provides site-level data for single sites or aggregated across portfolios of sites. Historical data can be reviewed in a graphing interface, providing visibility into site-level behavior and allowing those data to be exported.

Access to Powerhub is included for all Megapacks. For more information, see the *Powerhub User Manual*.

The Powerhub API provides programmatic access to the same site-level data available via Powerhub, with the ability to send event-based controls as well. Purchasing access to the Powerhub API includes controls capabilities via the Powerhub user interface.

For questions about Powerhub or the Powerhub API, contact powerhub_support@tesla.com.



Revision History

Revision	Date	Description
1.8	May 13, 2022	- Updated round-trip efficiency section, including graphs (Round-Trip Efficiency and Energy Consumption on page 5) - Simplified roof live-load language (Roof Loading on page 18)
1.7	April 6, 2022	- Modified 4-hour energy rating (Power and Energy on page 5) - Modified Voltage Ride Through settings (Voltage Ride Through on page 11) - Modified audible noise specification (Audible Noise on page 20) - Updated kVA increments from 100 kVA to 50 kVA (Inverter Configurations on page 13) - Modified enclosure dimensions (Dimensions and Mass on page 19) - Minor updates to product language and definitions (System Description on page 2) - Updated nominal ratings with max configurable power to have time dependency (Power and Energy on page 5) - Modified round-trip efficiency figures and added more detailed energy consumption figures (Round-Trip Efficiency and Energy Consumption on page 5) - Updated battery module configurations; increased minimum allowable product configuration to 400 kVA (Battery Module Configurations on page 15, Inverter Configurations on page 13) - Improved language and made minor updates to environmental specifications (Electrical Specifications on page 5) - Updated Roof Loading on page 18 and Solar Loading on page 18 - Provided additional transportation details (Transportation on page 19) - Created installation section (Installation on page 19) - Provided additional communication and control information (Communication and Control on page 21)
1.6.4	November 9, 2021	- Real power / energy ratings update - RTE update (Round-Trip Efficiency and Energy Consumption on page 5) - Audible noise spec update (Audible Noise on page 20)
1.6.3	July 21, 2021	Updated unit height
1.6.2	07-14-2021	Ratings & RTE update
1.6.1	05-04-2021	- Ratings update - New language in inverter section



Revision	Date	Description
		• Battery modules depopulation
1.6	04-15-2021	• Ratings update to reflect additional battery modules
1.51	04-06-2021	• Product name is now officially Megapack 2
1.5	03-30-2021	• Slightly increased unit width • 2hr and 4hr ratings update
1.4.1	03-05-2021	• Updated product long-term RTE
1.4	02-23-2021	• Updated product mass projection • Updated product RTE
1.3	02-02-2021	• Rating update
1.2	11-25-2020	• Rating update
1.1.1	10-02-2020	• Fixed a broken cross-reference link
1.1	09-30-2020	• First version of specification for limited external release
1.0	09-14-2020	• Initial release candidate for external version of next-generation Megapack specification



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